PROBLEM CL18

Liquid Ejector: Entrained Flow Rate, Outlet Velocity and Heating

Original statement · CL18.pdf

Ejector Bernoulli Mixing Dissipation

OFFICIAL PROBLEM STATEMENT

The ejector shown in the figure of CL18.pdf is used to discharge a container filled with a perfect liquid of density ρ and constant specific heat c into the atmosphere, also at pressure p_a. A high-velocity jet of the same liquid, with velocity U_j and cross section A_j, is injected along the axis of the discharge pipe, whose cross section is A. The jet entrains fluid as it develops, producing an underpressure at the injection section and drawing the fluid from the container. Mixing of the jet with the surrounding liquid is completed downstream, so that at the outlet the liquid discharges as a uniform jet with velocity U_o. For a given the geometry (i.e., given values of A_j and A), we want to compute the characteristics of the flow in terms of the injection velocity U_j. In the computation, assume that the effect of gravity is negligible. Address in order the following questions:

- Assuming that the acceleration of the fluid upstream from the injector is steady and inviscid, write down the equation that relates the ambient pressure p_a with the velocity and pressure at the entrance of the pipe, U_e and p_e.
- Apply the continuity and momentum balance equations to the control volume indicated in the figure. Simplify the result by assuming that the effect of friction on the wall surface is negligibly small.
- Use the previous equations to show that the volumetric discharge flow rate is given by $Q = U_e(A - A_j) = \dfrac{U_j(A - A_j)}{1 + A/\sqrt{2A_j(A - A_j)}}$ and that the velocity at the outlet satisfies $\dfrac{U_o}{U_j} = \dfrac{A_j + \sqrt{2A_j(A - A_j)}}{A + \sqrt{2A_j(A - A_j)}}$.
- Assuming that the values of the temperatures T_e and T_j at the entrance section are equal to the ambient value T_a, obtain the temperature T_o at the outlet section when the pipe walls are adiabatic.

Source: CL18.pdf (page 1, midterm of Nov 11, 2013). Notes.pdf: Bernoulli's equation Eq. 4.17, integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible liquid, $e = cT$; neglect $\bar{g}$; friction on the pipe wall negligible; adiabatic walls.
- Entrance section (area A): the jet (A_j, U_j) and the entrained liquid (annulus $A - A_j, U_e$) are both at the pressure p_e (parallel streams). Outlet: uniform U_o at p_a (complete mixing).
- Data: $[\rho,\ c,\ A,\ A_j,\ U_j,\ p_a,\ T_a]$. Unknowns: $[U_e,\ p_e,\ U_o,\ Q,\ T_o]$.

Symbol	Meaning	SI unit
A, A_j	pipe and jet areas	m^2
U_j, U_e, U_o	jet, entrained and outlet velocities	m/s
p_e	pressure at the entrance section	Pa
Q	entrained volume flow rate	m^3/s

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the pipe; $\bar{e}_x$ along the pipe axis in the flow direction. Fixed control volume between the entrance and the outlet sections.
- Surfaces: Σ_e (entrance, $\bar{n} = -\bar{e}_x$, split into the jet A_j and the annulus $A - A_j$), Σ_o (outlet, $\bar{n} = +\bar{e}_x$), Σ_w (straight pipe wall, $\bar{n} \perp \bar{e}_x$).

PHASE 3

Balances, Surface by Surface

1. Bernoulli upstream of the entrance

From the still liquid of the container (at p_a) to the entrance (Notes Eq. 4.17):

$$\boxed{p_a = p_e + \tfrac{1}{2}\rho U_e^2} \tag{1}$$

2. Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho \, dV}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \, (\bar{v} - \underbrace{\bar{v}_c}_{0}) \cdot \bar{n} \, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \, \bar{v} \cdot \bar{n} \, d\sigma$
Σ_e (annulus)	$-\rho U_e (A - A_j)$
Σ_e (jet)	$-\rho U_j A_j$
Σ_o	$+\rho U_o A$
Σ_w	0

$$\boxed{U_o A = U_e (A - A_j) + U_j A_j} \tag{2}$$

2. Linear momentum, x component (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho \bar{v} \, dV}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \bar{v} \, [(\bar{v} - \underbrace{\bar{v}_c}_{0}) \cdot \bar{n}] \, d\sigma = -\int_{\Sigma_c} (p - p_a) \, \bar{n} \, d\sigma + \int_{\Sigma_c} \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma + \underbrace{\int_{V_c} \rho \bar{g} \, dV}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho v_x\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,n_x\,d\sigma$	$\int (\bar{\bar{\tau}}'\cdot\bar{n})_x\,d\sigma$
Σ_e (annulus)	$-\rho U_e^2 (A - A_j)$	$(p_e - p_a)(A - A_j)$	0
Σ_e (jet)	$-\rho U_j^2 A_j$	$(p_e - p_a)A_j$	0
Σ_o	$+\rho U_o^2 A$	0 ($p = p_a$)	0
Σ_w	0	0 ($n_x = 0$)	0 (friction neglected)

With (1), $(p_e - p_a)A = -\frac{1}{2}\rho U_e^2 A$.

$$\boxed{U_o^2 A = U_e^2\left(\frac{A}{2} - A_j\right) + U_j^2 A_j} \tag{3}$$

3. Flow rate and outlet velocity

Squaring (2) and dividing by A, then equating to (3):

$$\left[U_e(A - A_j) + U_j A_j\right]^2 = A\left[U_e^2\left(\frac{A}{2} - A_j\right) + U_j^2 A_j\right].$$

Expanding, $U_e^2\left[(A-A_j)^2 - \frac{A^2}{2} + AA_j\right] + 2U_eU_jA_j(A-A_j) + U_j^2(A_j^2 - AA_j) = 0$. With $k = A_j(A - A_j)$ and $x = U_e/U_j$, the bracket is $\frac{A^2}{2} - k$ and the equation reads

$$\left(\frac{A^2}{2} - k\right)x^2 + 2kx - k = 0 \implies x = \frac{-k \pm \sqrt{k^2 + k\left(\frac{A^2}{2} - k\right)}}{\frac{A^2}{2} - k} = \frac{-k \pm \frac{As}{2}}{\frac{(A-s)(A+s)}{2}}, \qquad s \equiv \sqrt{2k} = \sqrt{2A_j(A - A_j)}.$$

The $+$ sign gives $x = \frac{s(A-s)}{(A-s)(A+s)} = \frac{s}{A+s}$; the $-$ sign gives $x = -\frac{s}{A-s} < 0$ (liquid flowing back into the container, rejected). Hence

$$\frac{U_e}{U_j} = \frac{1}{1 + A/\sqrt{2A_j(A - A_j)}}, \qquad \boxed{Q = U_e(A - A_j) = \frac{U_j(A - A_j)}{1 + A/\sqrt{2A_j(A - A_j)}}} \tag{4}$$

and from (2), $\frac{U_o}{U_j} = \frac{x(A-A_j)+A_j}{A} = \frac{s(A-A_j)+A_j(s+A)}{A(A+s)} = \frac{sA+A_jA}{A(A+s)}$:

$$\boxed{\frac{U_o}{U_j} = \frac{A_j + \sqrt{2A_j(A-A_j)}}{A + \sqrt{2A_j(A-A_j)}}} \tag{5}$$

4. Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_e} \rho\left(e + \frac{v^2}{2}\right)[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\, d\sigma = -\int_{\Sigma_e} p\,\bar{v}\cdot\bar{n}\, d\sigma + \int_{\Sigma_e} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\, dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_e} \bar{q}\cdot\bar{n}\, d\sigma + \underbrace{\int_{V_c} Q_R\, dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\, d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\, d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$	$-\int \bar{q}\cdot\bar{n}\, d\sigma$
Σ_e (annulus)	$-\rho U_e(A-A_j)\left(cT_e + \frac{U_e^2}{2}\right)$	$+p_e U_e(A-A_j)$	0	0
Σ_e (jet)	$-\rho U_j A_j\left(cT_j + \frac{U_j^2}{2}\right)$	$+p_e U_j A_j$	0	0
Σ_o	$+\rho U_o A\left(cT_o + \frac{U_o^2}{2}\right)$	$-p_a U_o A$	0	0
Σ_w	0	0	0	0 (adiabatic)

Pressure work: by (2) and (1), $p_e[U_e(A-A_j) + U_jA_j] - p_aU_oA = (p_e - p_a)U_oA = -\frac{1}{2}\rho U_e^2 U_o A$. With $T_e = T_j = T_a$ the internal-energy terms give $\rho c U_o A(T_o - T_a)$; using $U_e(A-A_j) = U_oA - U_jA_j$ and dividing by $\rho U_o A U_j^2$:

$$\boxed{\frac{c\,(T_o - T_a)}{U_j^2} = \frac{U_jA_j}{2U_oA}\left[1 - \left(\frac{U_e}{U_j}\right)^2\right] - \frac{1}{2}\left(\frac{U_o}{U_j}\right)^2} \tag{6}$$

Erratum in the official solution

CL18.pdf writes the energy balance as in the table but boxes $\frac{c(T_o-T_a)}{U_j^2} = \frac{U_j A_j}{U_o A}\left[1-\left(\frac{U_e}{U_j}\right)^2\right] - \left(\frac{U_o}{U_j}\right)^2$, without the two factors $\frac{1}{2}$ (all kinetic energies are $U^2/2$), i.e. twice the value of (6). Check: for $A = 1\ \mathrm{m}^2$, $A_j = 0.1\ \mathrm{m}^2$, $U_j = 10$ m/s, $c = 4186$ J/(kg K), the residual of the energy balance (per unit density) is 0 with (6) and 20.6 W/(kg/m^3) with the boxed expression. The computed value is kept.

PHASE 4

Interpretation, Limits and Dimensional Check

- $T_o > T_a$: the mixing of the fast jet with the slow entrained liquid dissipates kinetic energy, which appears as heat. The rise is tiny for a liquid (c is large).
- Limit $A_j \to 0$: $s \to 0$, $Q \to 0$ and $U_o \to 0$ (no jet, no pumping). Limit $A_j \to A$: $s \to 0$, $Q \to 0$ and $U_o \to U_j$ (the jet fills the pipe).
- Dimensions (SI): $[Q] = (\mathrm{m/s})\,\mathrm{m}^2 = \mathrm{m}^3/\mathrm{s}$; $[cT/U^2] = 1$.
- Numerical illustration (hypothetical data): $A = 1\ \mathrm{m}^2$, $A_j = 0.1\ \mathrm{m}^2$, $U_j = 10$ m/s, water ($c = 4186$ J/(kg K)) give $U_e = 2.9788$ m/s, $Q = 2.6809\ \mathrm{m}^3$/s, $U_o = 3.6809$ m/s and $T_o - T_a = 1.339$ mK; the residuals of (2) and (3) are below 10^{-14}.
- Agreement with the official solution: (1) to (5) as in CL18.pdf; (6) differs by the factors $\frac{1}{2}$ discussed above.